

Robust QCar State Estimation with Analytical Consistency Gates and Learned Measurement Trust

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Abstract

Sensor corruption can degrade vehicle localization even when the nominal motion model is accurate. This paper presents a hybrid QCar estimator combining a calibrated physical predictor, explicit consistency checks, and a learned measurement-reliability model. A shared neural network with 337 parameters modulates diagonal analytical correction gains; it does not estimate an unrestricted Kalman gain matrix. Sensor availability, inferred reliability, and the simulator’s attack indicator are defined separately. Only corrupted sensor measurements, commands, and their causal history enter online inference. Training uses simulated drives with known corruption and state targets, while evaluation uses continuous trajectories with disjoint drive seeds and no intermediate truth resets. Across 13 nominal clean/fault conditions, the estimator has lower position RMSE than the repository EKF. For GPS jumps, pooled position RMSE decreases from 0.7308 to 0.0139 m; for gradual GPS drift it decreases from 0.5975 to 0.0701 m. A neural ablation obtains 0.0142 and 0.2202 m, respectively, showing that analytical design explains most jump rejection, whereas learning contributes more to gradual drift. Separate Stanley–PID control experiments assess the effect on driving. A combined GPS/tachometer attack under calibration mismatch increases hybrid error to 0.4638 m, above the analytical ablation’s 0.4019 m. The results establish an empirical improvement in the tested simulator, with explicit limits on model mismatch, learning attribution, and transfer to physical vehicles.

Keywords: vehicle state estimation; sensor faults; measurement reliability; hybrid filtering; QCar simulation; causal evaluation.

1 Introduction

A vehicle estimator must reject unreliable sensor information without losing the motion information required by its controller. This is more demanding than smoothing nominal measurement noise. A biased or frozen sensor can remain syntactically valid, and its repeated corrections can move the estimate until the corrupted value becomes consistent with the estimate itself. A corrupted tachometer or inertial signal can also affect prediction before a position update is considered.

KalmanNet combines a structural state-space model with recurrent neural gain computation for partially known dynamics [1]. RobustStateNet models sensor branches with recurrent networks and selective feature fusion within a prediction/update architecture [2]. These works motivate combining physical structure and learned reliability, but they do not imply that a particular local implementation is robust to the QCar fault distribution. The method studied here retains the repository name **RobustKLNet**, with the backend identifier **innovation_trust**. Its active architecture is a physical predictor with analytical diagonal gains and learned trust, rather than the earlier GRU gain network.

The contribution is an implemented and evaluated design for this simulation setting: (i) consistency checks protect both prediction and correction; (ii) analytical rejection and learned reliability are combined explicitly; and (iii) continuous estimation and feedback-control experiments separate overall performance from the incremental contribution of learning. The study is an empirical engineering evaluation. It does not claim a new general optimal filter, a formal attack-identification guarantee, or superiority to all robust Kalman-filter variants.

2 Estimation problem and information available online

The state and direct measurement vector are

$$\mathbf{x}_k = [x_k, y_k, \psi_k, v_k]^\top, \quad \mathbf{z}_k = [x_k^{\text{GPS}}, y_k^{\text{GPS}}, \psi_k^{\text{GPS}}, v_k^{\text{tach}}]^\top. \quad (1)$$

The online inputs additionally include throttle u_k , the steering signal δ_k , yaw rate ω_k^{IMU} , longitudinal acceleration a_k^{IMU} , and the sample interval Δt_k . Angular differences are wrapped to $[-\pi, \pi)$. GPS heading is checked

before it can contaminate a separate heading-fusion stage. The tachometer is one physical source; four repeated copies are not treated as independent wheel observations.

For an available channel, a useful fault notation is

$$z_{k,j} = h_j(\mathbf{x}_k) + n_{k,j} + d_{k,j}, \quad (2)$$

where $d_{k,j}$ denotes corruption. Replacement and freeze faults can be represented by a time-dependent $d_{k,j}$; dropout instead removes the observation. The estimator does not observe $d_{k,j}$, the clean counterpart, the true moving state, or the scheduled attack interval. Its initialization uses the known spawn pose once.

Table 1: Three distinct quantities: attack annotation, availability, and inferred trust.

Quantity	Values	Meaning and use
Attack indicator A_k	$\{0, 1\}$	Known only to the experiment generator and scorer; 1 marks an injected fault interval. It is not an inference input.
Availability $a_{k,j}$	$\{0, 1\}$	Obtained from sensor validity/freshness metadata and finite-value checks. A corrupted observation may still have $a_{k,j} = 1$.
Applied trust $\gamma_{k,j}$	$[0, 1]$	Computed online. A small value suppresses that measurement’s correction; it is not the attack signal or a calibrated probability of malicious activity.

3 Hybrid estimator

3.1 Calibrated prediction and protection of motion inputs

Let $F(v, u, \Delta t)$ be the calibrated throttle-to-speed propagation. For the evaluated lookup model,

$$\begin{aligned} v_{ss}(u) &= \text{interp}(u; \mathcal{U}, \mathcal{V}), & \dot{v}_0 &= \frac{v_{ss}(u) - v}{\tau}, \\ \dot{v} &= \text{clip}\left(\dot{v}_0, -2, \frac{2v_{sw}}{\max(v, v_{sw})}\right), & F(v, u, \Delta t) &= \text{clip}(v + \Delta t \dot{v}, -2.5, 2.5). \end{aligned} \quad (3)$$

The nominal constants are $\tau = 0.301$ s, $v_{sw} = 0.1$ m/s and wheelbase $L = 0.26$ m. Acceleration limits are in m/s^2 . A near-stop rule sets acceleration to zero when $|u| < 0.01$ and $|v| < 0.03$ m/s. Appendix A gives the static lookup points supplied as nominal calibration to both plant and estimators.

Two velocities are propagated: $v_k^- = F(\hat{v}_{k-1}, u_k, \Delta t_k)$ and a command-driven auxiliary velocity $v_k^{\text{cmd}} = F(v_{k-1}^{\text{cmd}}, u_k, \Delta t_k)$. The latter receives no direct tachometer correction. The tachometer’s influence on pose prediction is moderated by

$$\begin{aligned} q_k &= \exp\left[-\min\left\{\left(\frac{|v_k^{\text{tach}} - v_k^{\text{cmd}}|}{0.20}\right)^4, 60\right\}\right] \gamma_{k-1,v}, \\ v_k^{\text{use}} &= v_k^- + 0.65q_k(v_k^{\text{tach}} - v_k^-). \end{aligned} \quad (4)$$

The previous-trust factor is omitted at the first sample. A non-finite or detected frozen tachometer sets $q_k = 0$. Rejection therefore protects motion prediction as well as the velocity update.

A first-order, rate-limited steering state follows the steering input. Its bicycle yaw-rate prediction is compared with the IMU yaw rate and acceleration. The predictor uses the IMU rate when consistency tests accept it and otherwise uses the bicycle rate. Appendix B specifies these deterministic guards. With accepted yaw rate ω_k and midpoint speed $\bar{v}_k = (\hat{v}_{k-1} + v_k^{\text{use}})/2$, the prediction is

$$\mathbf{x}_k^- = \begin{bmatrix} \hat{x}_{k-1} + \bar{v}_k \Delta t_k \cos(\hat{\psi}_{k-1} + \omega_k \Delta t_k / 2) \\ \hat{y}_{k-1} + \bar{v}_k \Delta t_k \sin(\hat{\psi}_{k-1} + \omega_k \Delta t_k / 2) \\ \text{wrap}(\hat{\psi}_{k-1} + \omega_k \Delta t_k) \\ v_k^- \end{bmatrix}. \quad (5)$$

These equations describe the evaluated kinematic regime, not a learned tire/slip model.

3.2 Odometry reference and temporal features

An auxiliary state $\mathbf{o}_k$ is integrated using accepted speed and yaw rate, without the fast GPS position corrections. Its velocity is v_k^{cmd} . Persistent GPS displacement can thus remain visible even after the main estimate begins moving toward that displacement. This reference is *not statistically independent*: it shares motion inputs and accepted speed history with the main estimate.

The innovation and reference discrepancy are

$$r_{k,j} = z_{k,j} - x_{k,j}^-, \quad d_{k,j}^o = z_{k,j} - o_{k,j}, \quad (6)$$

with angular wrapping in channel ψ . Successive available observations and their corresponding reference states give a motion-disagreement feature $\Delta z_j - \Delta o_j$. An exponentially weighted discrepancy is updated on available samples as $E_{k,j} = 0.85E_{k-1,j} + 0.15d_{k,j}^o$. Unavailable channels retain their history.

Each channel has a 12-dimensional feature vector

$$\mathbf{f}_{k,j} = \begin{bmatrix} \log(1 + |r_{k,j}|/s_j) \\ \log(1 + |d_{k,j}^o|/s_j) \\ \log(1 + |\Delta z_j - \Delta o_j|/s_j) \\ \log(1 + |E_{k,j}|/s_j) \\ \min(T_{k,j}^{\text{repeat}}, 3) \\ |v_k^{\text{use}}| \\ |\omega_k| \\ \min(\Delta t_k^{\text{GPS}}, 2) \\ \mathbf{e}_j \end{bmatrix}, \quad \mathbf{s} = [0.12, 0.12, 0.09, 0.10]^\top, \quad (7)$$

where $\mathbf{e}_j$ is a four-entry channel identity vector. T^{repeat} is an implementation-level repeated-value accumulator; its timing convention and the separate tachometer timer are stated in Appendix B. The features contain current measurements and past information only.

3.3 Analytical consistency and learned measurement trust

A shared MLP evaluates all four channels:

$$\rho_{k,j} = \text{sigmoid}(W_2 \text{ReLU}(W_1 \mathbf{f}_{k,j} + \mathbf{b}_1) + b_2). \quad (8)$$

Here $W_1 \in \mathbb{R}^{24 \times 12}$ and $W_2 \in \mathbb{R}^{1 \times 24}$, giving $288 + 24 + 24 + 1 = 337$ trainable parameters. Memory resides in the filter and its features; the MLP itself is not recurrent.

The deterministic consistency term is

$$\begin{aligned} e_{k,j} &= \max\{|r_{k,j}|, 0.75|d_{k,j}^o|\}, \\ b_{k,j} &= \exp[-\min\{(e_{k,j}/c_j)^4, 60\}], \quad \mathbf{c} = [0.30, 0.30, 0.20, 0.20]^\top. \end{aligned} \quad (9)$$

The entries of $\mathbf{c}$ are in metres, metres, radians and metres per second. Explicit frozen-sensor guards can set affected $b_{k,j}$ to zero. This rule strongly suppresses large discrepancies rather than repeatedly applying a fixed clipped residual. Availability, consistency and learning are combined:

$$\begin{aligned} \tilde{\gamma}_{k,j} &= a_{k,j} b_{k,j} \rho_{k,j}, & \gamma_{k,x} &= \gamma_{k,y} = \min(\tilde{\gamma}_{k,x}, \tilde{\gamma}_{k,y}), \\ \gamma_{k,\psi} &= \tilde{\gamma}_{k,\psi}, & \gamma_{k,v} &= \tilde{\gamma}_{k,v}. \end{aligned} \quad (10)$$

GPS position is treated as a pair. This avoids retaining one coordinate when the other indicates an unreliable position source. It does not assert statistical independence of heading, velocity and position errors.

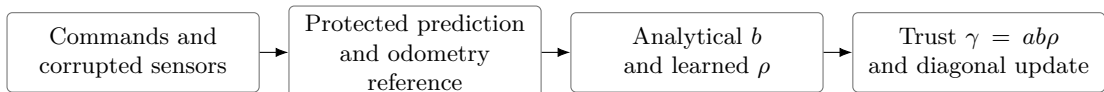


Figure 1: Online computation. Temporal states feed the next prediction and trust features. Attack indicators and clean references are absent from this inference path. The shared GPS-position minimum in Eq. (10) precedes correction.

3.4 Analytical correction and its limited guarantee

Four positive variance-like scalars determine the gains:

$$\begin{aligned} P_{k,j}^- &= P_{k-1,j}^+ + Q_j \frac{\Delta t_k}{0.02}, & K_{k,j} &= \frac{P_{k,j}^-}{P_{k,j}^- + R_j}, \\ \alpha_{k,j} &= K_{k,j} \gamma_{k,j}, & \hat{x}_{k,j} &= x_{k,j}^- + \alpha_{k,j} r_{k,j}, \\ P_{k,j}^+ &= \max\{(1 - \alpha_{k,j})P_{k,j}^-, 10^{-9}\}. \end{aligned} \quad (11)$$

Heading is wrapped after correction. Appendix A specifies the initial variances, Q_j and R_j . The recursion omits Jacobian-based cross-covariance propagation and uses P as a gain-scheduling quantity; it is not established as a calibrated posterior covariance or an optimal EKF recursion.

Since $P^- > 0$, $R > 0$ and $\gamma \in [0, 1]$, one has $0 \leq \alpha < 1$. For a non-angular channel the direct correction is a convex interpolation between prediction and measurement; for heading the same property holds along the wrapped residual. The diagonal update prevents a position residual from *directly* correcting heading or velocity in that update. This local algebraic property does not prove bounded trajectory error, attack detection, or global stability. Subsequent prediction still couples the states.

After correction, each reference pose coordinate is slowly aligned only if $\gamma_{k,j} > 0.8$ and $|d_{k,j}^o| < 0.65s_j$:

$$o_{k,j} \leftarrow o_{k,j} + \min(\Delta t_k^{\text{GPS}}/8, 0.1)d_{k,j}^o, \quad j \in \{x, y, \psi\}. \quad (12)$$

This reduces nominal drift while limiting rapid assimilation of suspicious GPS offsets. Coherent slow corruption can nevertheless contaminate the shared reference.

3.5 Relationship to the previous implementation

The retained legacy implementation has a learned feature/measurement mask and a recurrent gain calculation. The active backend replaces that recurrent gain with Eq. (11), uses the physical predictor of Eq. (5), and learns only ρ . It also checks GPS heading before fusion, rejects frozen tachometer influence in prediction, and disables publication of a clean-reference comparator as the estimator output. The persistent concept is prediction followed by reliability-weighted correction. The architectural change separates explicit safeguards, analytical gains and a smaller learned component.

4 Learning measurement reliability

4.1 Offline supervision

A training drive contains clean noisy sensor samples, corrupted samples and simulated true states. These are available to the training-data generator only. Let $\epsilon_{k,j}^{\text{fault}} = z_{k,j}^{\text{corrupt}} - z_{k,j}^{\text{clean}}$ with heading differences wrapped. Soft reliability targets are

$$y_{k,j} = \exp[-(|\epsilon_{k,j}^{\text{fault}}|/\eta_j)^2], \quad \boldsymbol{\eta} = [0.07, 0.07, 0.05, 0.06]^\top. \quad (13)$$

The two position targets are replaced by their minimum. A frozen value can retain high target reliability until it departs materially from the contemporaneous clean measurement. Unavailable measurements are excluded from the training examples. The generator’s binary attack labels are not the network inputs or the direct targets of this objective.

Training uses six drive seeds (1–6), and model selection uses two disjoint seeds (21–22). Drives last 30 s at 50 Hz. GPS rates cycle through 5, 10 and 20 Hz. The plant time constant varies by -8% , 0 , $+8\%$ and the steady-state speed curve by -4% , 0 , $+4\%$, according to the seed, while estimator calibration remains nominal. Each drive is replayed under the 13 conditions in Table 3, including clean data.

4.2 Corruption-weighted reliability and correction loss

The reliability term is weighted binary cross entropy:

$$\mathcal{L}_{\text{rel}} = -\frac{1}{M} \sum_i [1 + 3(1 - y_i)] [y_i \log \rho_i + (1 - y_i) \log(1 - \rho_i)]. \quad (14)$$

This is corruption weighting, not exact balancing by class frequency. The index i spans available channel observations.

A second term measures a candidate one-step correction on stored rollout features. For normalization scales $\ell = [0.04, 0.04, 0.02, 0.025]^\top$, define $\bar{r}_i = r_i/\ell_j$ and $\bar{e}_i = (x_i^- - x_i^{\text{true}})/\ell_j$, with angular wrapping. The data-collection filter stores

$$\bar{g}_i = \frac{K_i \gamma_i^{\text{collect}}}{\max(\rho_i^{\text{collect}}, 10^{-6})}, \quad \mathcal{L}_{\text{state}} = \frac{1}{M} \sum_i \min\{(\bar{e}_i + \rho_i \bar{g}_i \bar{r}_i)^2, 100\}. \quad (15)$$

The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{rel}} + 0.08 \mathcal{L}_{\text{state}}. \quad (16)$$

Collected gains, the reference state and GPS-position coupling are fixed while optimizing each minibatch. Equation (15) is thus a local surrogate, not differentiation through the full filter trajectory or through the current two-coordinate minimum. The clipping limits the influence of large state errors on this loss.

4.3 Two collection/training passes

The first pass collects features using the analytical filter with $\rho = 1$. After fitting the MLP, the second pass recollects causal features using the first learned checkpoint and continues optimization. This exposes training to states visited by a learned filter; it does not eliminate train/deployment distribution mismatch.

Optimization uses AdamW with learning rate 0.003 , weight decay 10^{-4} , batch size 4096 , gradient-norm clipping at 2 , and training seed 17 . Each pass permits 60 epochs. An improvement threshold of 10^{-5} and patience of 15 epochs govern early stopping; the best validation weights in each pass are restored. Both recorded passes ran 60 epochs. The selected epochs in Table 2 use one-based numbering. A lower second-pass loss is descriptive because its collected examples differ from the first pass; the external continuous benchmark determines estimation quality.

Table 2: Recorded losses at the selected epoch of each collection/training pass.

Pass	Selected epoch	Training loss	Validation loss
1	60	0.07792	0.05920
2	58	0.04374	0.03752

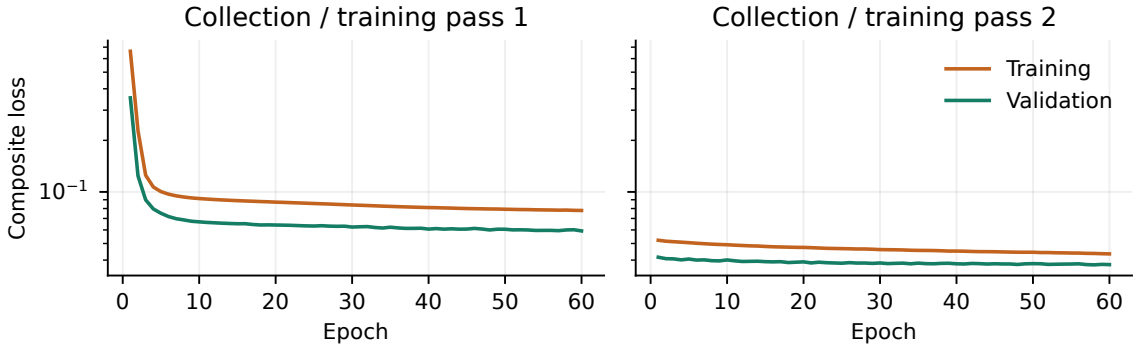


Figure 2: Optimization histories from saved checkpoint metadata. The neural network is trained on channel examples collected from causal rollouts; it is not an online optimizer during evaluation.

5 Experimental protocol

5.1 Plant, sensors, and causal evaluation

Experiments use the repository’s `MockQCar` plant, also used by `fake_vehicle_real_logic.py`. The main benchmark uses its kinematic model, calibrated throttle lookup and default steering actuator. Throttle and steering commands vary sinusoidally with seed-dependent phases; a short stop occupies approximately 72–78% of each drive. The electronics twin is disabled in these benchmarks.

The nominal test comprises seeds 101–103, 60 s per drive, $\Delta t = 0.02$ s, and 10 Hz GPS. Independent added sensor noise has standard deviations 0.035 m per GPS coordinate, 0.015 rad for GPS heading, 0.012 m/s for tachometer speed, 0.008 rad/s for gyro rate, and 0.04 m/s² per acceleration component. Initial positions are zero, initial headings are seed-dependent, and speed starts at zero. Each estimator receives the same corrupted samples and one known initial pose. Its state remains continuous for all 3000 steps; no target reset or overlapping-window averaging is performed.

The test seeds do not overlap training/validation seeds. However, the fault families and command-generation family are represented during training. These are tests on new drives within specified families, not demonstrations on previously unseen attack mechanisms. The saved seeds should not be reused as an untouched final test after further model tuning.

5.2 Fault conditions

For each non-clean open-loop condition, fault intervals are $[0.22T, 0.45T)$ and $[0.56T, 0.70T)$: $[13.2, 27.0)$ and $[33.6, 42.0)$ s for $T = 60$ s. Each interval samples a sign $\varsigma \in \{-1, 1\}$ and jump magnitude $B \sim U(0.65, 1.4)$ m. The schedule is external to the estimator.

Table 3: Nominal open-loop conditions. Offsets are applied to estimator inputs after plant simulation; steering corruption does not alter the plant command in this test. t_a is time since the current fault interval began.

Condition	Injected change
Clean	Nominal sensor noise only.
GPS jump	Position offsets $(\varsigma B, -0.7\varsigma B)$ m.
GPS ramp	Position offsets $(0.12\varsigma t_a, -0.08\varsigma t_a)$ m.
GPS freeze	Reuse the first GPS packet in the interval, including heading.
GPS dropout	Remove GPS packets, including GPS heading.
GPS noise	Add independent position noise with standard deviation 0.65 m.
Heading bias	Add 0.9ς rad to GPS heading, then wrap.
Tachometer bias	Add 0.65ς m/s to measured speed.
Tachometer scale	Multiply measured speed by 1.9.
Tachometer freeze	Hold the speed measurement at interval onset.
IMU bias	Add 0.8ς rad/s to yaw rate and 1.5ς m/s ² to each of the first two acceleration components.
Steering bias	Add 0.3ς rad to the estimator’s steering input.
GPS + tachometer	Combine GPS jump and tachometer bias.

5.3 Comparators and metrics

Four methods are evaluated: the repository fallback EKF, the retained legacy GRU checkpoint, the analytical ablation, and the full hybrid. The EKF is configured with `use_qcar_ekf=False`. All receive the same nominal wheelbase and throttle-model calibration; their internal prediction, noise handling and correction structures still differ. The legacy model is run continuously with a one-step streaming sequence and is not retrained in this study. Its results describe that checkpoint and adapter configuration, not the potential of all recurrent architectures. The analytical ablation sets $\rho = 1$ while retaining the physical predictor, analytical gates, temporal memory and other safeguards. It is the principal learning-attribution comparison.

For N samples, position RMSE is

$$\text{RMSE}_p = \sqrt{\frac{1}{N} \sum_{k=1}^N [(\hat{x}_k - x_k)^2 + (\hat{y}_k - y_k)^2]}. \quad (17)$$

Heading RMSE uses wrapped error; speed RMSE is in m/s. For equal-length drives, reported pooled RMSE is $\sqrt{S^{-1} \sum_{s=1}^S \text{RMSE}_s^2}$. Tables report complete-drive error, including attack and recovery. Additional figures report error only during injection and during the first 3 s following each injection episode. The latter is a fixed post-attack window, not a measured settling time or proof of recovery. Position tails and maxima remain in the saved JSON. With only three nominal seeds and two stress seeds, no significance or population-level confidence claim is made.

6 Results

6.1 Continuous state estimation

Table 4 reports lower position RMSE for the hybrid than the EKF in all 13 nominal conditions. Clean RMSE is 0.0120 m, versus 0.0217 m for the EKF. The largest EKF position error occurs with frozen GPS, at 1.9428 m; the hybrid gives 0.0198 m. GPS dropout increases reliance on propagation, with errors of 0.0724 and 0.0232 m, respectively. These results concern the stated calibration and attack durations.

Table 4: Complete-drive position RMSE (m), pooled over three 60 s drives per condition. Analytical disables only the trust MLP; Hybrid is the current RobustKLNet backend.

Condition	EKF	Legacy GRU	Analytical	Hybrid
Clean	0.0217	0.7342	0.0123	0.0120
GPS jump	0.7308	0.9750	0.0142	0.0139
GPS ramp	0.5975	0.9206	0.2202	0.0701
GPS freeze	1.9428	2.4595	0.0255	0.0198
GPS dropout	0.0724	4.3690	0.0234	0.0232
GPS noise	0.2574	0.2297	0.0540	0.0166
Heading bias	0.0896	0.9627	0.0128	0.0124
Tachometer bias	0.0851	1.2780	0.0123	0.0119
Tachometer scale	0.0808	1.1376	0.0207	0.0119
Tachometer freeze	0.0366	1.2176	0.0123	0.0119
IMU bias	0.0218	0.7362	0.0123	0.0120
Steering bias	0.0222	1.0482	0.0123	0.0120
GPS + tachometer	0.7565	1.4769	0.0139	0.0136

Table 5 shows that improvement is not restricted to position. A GPS heading bias produces 0.5422 rad heading RMSE for the EKF and 0.0070 rad for the hybrid. Tachometer bias produces 0.3705 m/s speed RMSE for the EKF and 0.0046 m/s for the hybrid. The analytical ablation obtains similar performance in several of these conditions, consistent with the explicit motion-input guards and measurement rejection.

Table 5: Heading RMSE (rad) and speed RMSE (m/s), pooled over the same three drives. Analytical and hybrid results reveal the incremental effect of learning.

Condition	ψ : EKF	Analyt.	Hybrid	v : EKF	Analyt.	Hybrid
Clean	0.0089	0.0058	0.0058	0.0075	0.0058	0.0058
GPS jump	0.0090	0.0058	0.0058	0.0075	0.0058	0.0058
GPS ramp	0.0089	0.0058	0.0058	0.0075	0.0058	0.0058
GPS freeze	1.0118	0.0080	0.0067	0.0075	0.0058	0.0058
GPS dropout	0.0261	0.0070	0.0070	0.0075	0.0058	0.0058
GPS noise	0.0095	0.0058	0.0058	0.0075	0.0058	0.0058
Heading bias	0.5422	0.0070	0.0070	0.0075	0.0058	0.0058
Tachometer bias	0.0245	0.0058	0.0058	0.3705	0.0046	0.0046
Tachometer scale	0.0238	0.0058	0.0058	0.3526	0.0261	0.0046
Tachometer freeze	0.0122	0.0058	0.0058	0.1334	0.0049	0.0048
IMU bias	0.0221	0.0058	0.0058	0.0075	0.0058	0.0058
Steering bias	0.0494	0.0058	0.0058	0.0075	0.0058	0.0058
GPS + tachometer	0.0247	0.0058	0.0058	0.3705	0.0046	0.0046

Figure 3 makes the comparison visible across all conditions and all three state-error measures. Each point is a pooled RMSE; each segment spans the smallest and largest of the three per-drive RMSEs. These segments show observed seed variation, not confidence intervals. The logarithmic scale accommodates the retained GRU’s larger errors without removing that comparator from the display.

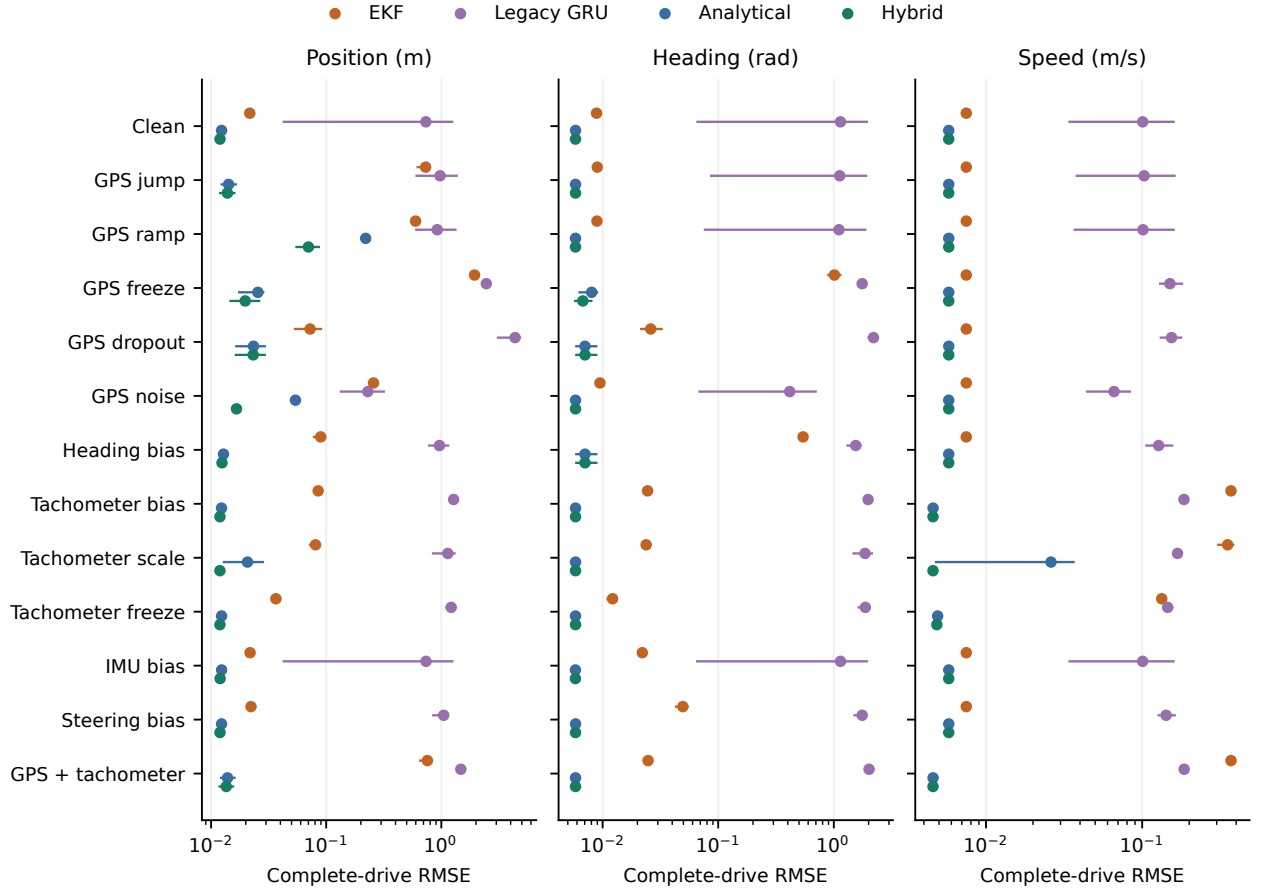


Figure 3: Complete nominal benchmark: three seeds, 13 conditions and four estimators. Dots show pooled RMSE and horizontal segments show the per-drive minimum–maximum range. All panels use logarithmic error axes; lower is better. The legacy GRU is the retained checkpoint, not an equally retrained architecture comparison.

6.2 What learning contributes

The ablation is essential for interpreting robustness. For GPS jumps, removing learning changes position RMSE only from 0.0139 to 0.0142 m. Large jump rejection is therefore predominantly explained by analytical design in this comparison. For GPS ramp corruption, learning reduces error from 0.2202 to 0.0701 m, approximately 68.2%. Under increased GPS noise the reduction is 0.0540 to 0.0166 m, approximately 69.2%. Tachometer scaling also benefits: speed RMSE decreases from 0.0261 to 0.0046 m/s. Figure 4 separates improvement over the EKF from improvement over the analytical ablation. An entry of 0.10 means one tenth of the reference RMSE, whereas 1.10 means 10% more error. Values close to one indicate little incremental benefit, even if both methods substantially outperform the EKF.

Figure 5 illustrates the trust factors for three corruptions of one drive. Learned reliability and analytical consistency are distinct quantities; their combination determines applied trust. The shaded attack intervals are annotations supplied by the experiment for interpretation. Online trust is inferred from sensor/model disagreement, not copied from those annotations. The diagnostic replays are checked against the saved state trajectories before the figures are generated. A low learned weight by itself is not the complete rejection mechanism: availability, the analytical factor and GPS-coordinate coupling also matter.

6.3 Observed masks, gains and observer outputs

Figure 6 displays the actual four-channel applied mask, together with GPS availability. A value near one permits a measurement correction; a value near zero suppresses it. This is a reliability weight, not an attack probability or the simulator’s attack label. The heatmaps are sampled at the nominal GPS schedule, including missing GPS slots. Plotting every 50 Hz step without this distinction would interleave ordinary GPS non-arrival with rejected GPS and make a healthy asynchronous sensor look faulty. The velocity row is sampled at the same display times, although it updates at 50 Hz.

The dropout panel illustrates a separate mechanism: $a^{GPS} = 0$ forces pose trust to zero even without classifying an attack. A GPS position jump instead arrives with $a^{GPS} = 1$; rejection follows from consistency and learned reliability. Equality of the x and y rows follows from the implemented position-pair minimum. Heading and speed are separately weighted, subject to their own evidence and guards.

The raw diagonal gain K_j and the applied gain $\alpha_j = K_j \gamma_j$ must be distinguished. Rejected updates can increase the variance-like quantity P_j^- and therefore increase raw K_j while the applied gain remains near zero. A large raw gain alone is consequently not evidence that an attack enters the state. Figure 7 connects gain modulation to the actual correction and resulting state error. For the hybrid, the plotted correction is $|\alpha_j r_j|$; for the EKF it is $|(K^{EKF} r^{EKF})_j|$, including cross-channel terms. All methods use their own prediction and residual. The comparison describes complete observers, not the isolated effect of replacing one gain while holding all other states fixed.

Figure 8 provides a complementary matrix view at the first attacked GPS update. Since the state components have different units, the display uses $D^{-1}KD$, where $D = \text{diag}(0.12, 0.12, 0.09, 0.10)$ contains the fixed feature scales. For the hybrid and ablation, K in this display means their effective correction matrix unless explicitly labeled raw. This normalization makes matrix entries dimensionless; it does not make the observers’ priors or noise assumptions identical. The matrix view exposes which residual can directly change which state component. It does not quantify indirect coupling through subsequent predictions.

Figure 9 then shows the four state outputs under simultaneous GPS and tachometer corruption. Received measurements, simulator truth and observer outputs are plotted together, and errors are shown separately so visually overlapping curves can still be compared. The retained GRU is included in the error panels; omitting its state excursions from the left-hand panels preserves the resolution of the other outputs. The aggregate comparison in Fig. 3 includes all methods.

6.4 Attack-period and post-attack behavior

Complete-drive averages alone can conceal a large error during corruption or delayed recovery after its removal. Figure 10 therefore recomputes the same state metrics using only attack samples and, separately, only the first 3 s after each attack. For these 60 s drives the post-attack windows are [27, 30) and [42, 45) s. The attack-only values reproduce the corresponding saved benchmark metrics. The post-attack values are newly derived from the saved trajectories without rerunning or retuning a filter. All three test seeds contribute to every reported ratio. The hybrid has lower pooled error than the EKF in every displayed attack-period and post-attack cell. This is evidence for these specified windows and fault conditions; it is not a guarantee for arbitrary attack durations or recovery behavior.

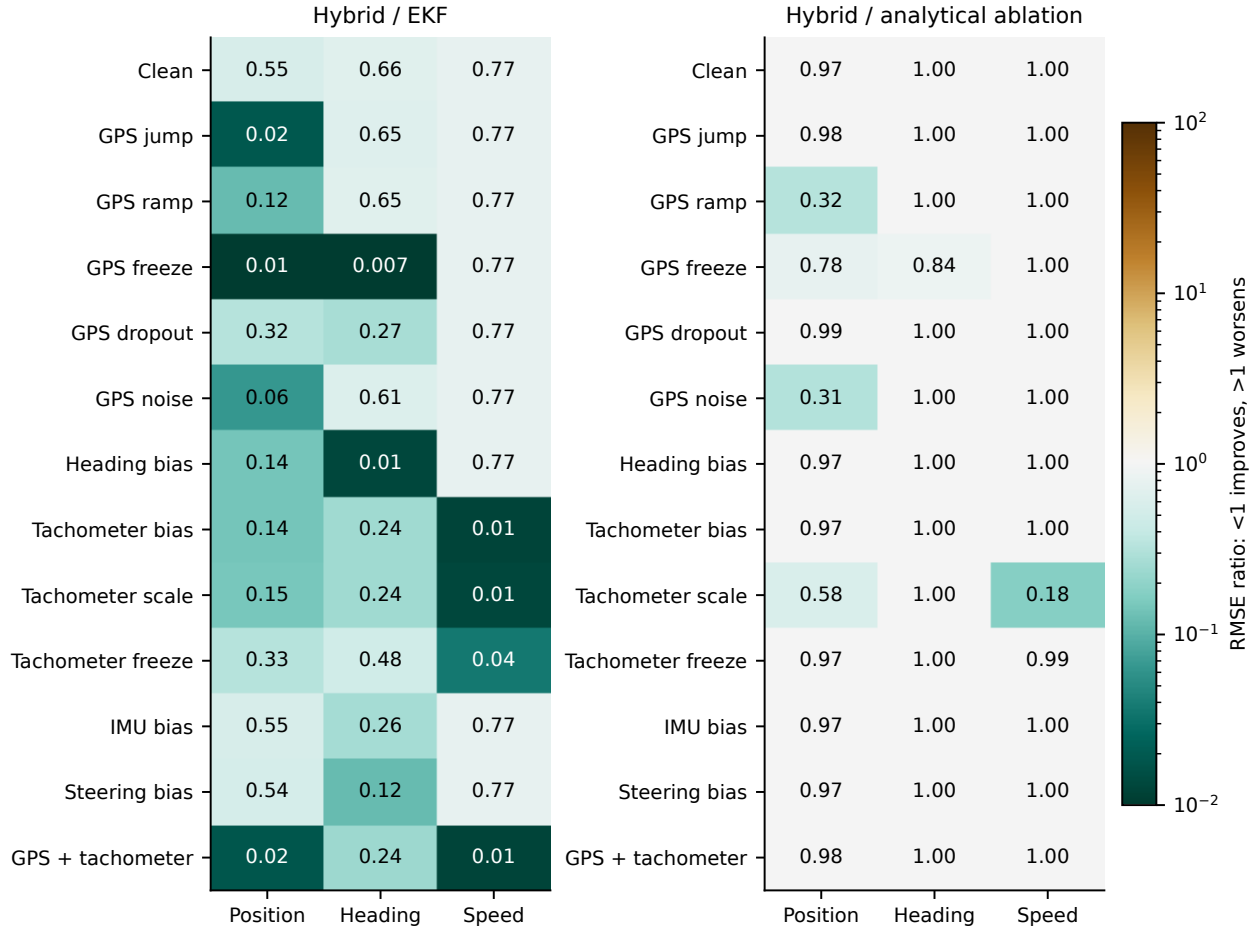


Figure 4: Ratios of pooled nominal RMSE. Left: hybrid relative to EKF. Right: hybrid relative to the analytical ablation. Values below one favor the hybrid; values above one favor the comparator. Numerical labels give the ratios without color-scale clipping. The logarithmic color scale is centered at equality and saturates outside $[0.01, 100]$. Learning’s incremental effect is evaluated in the right-hand panel.

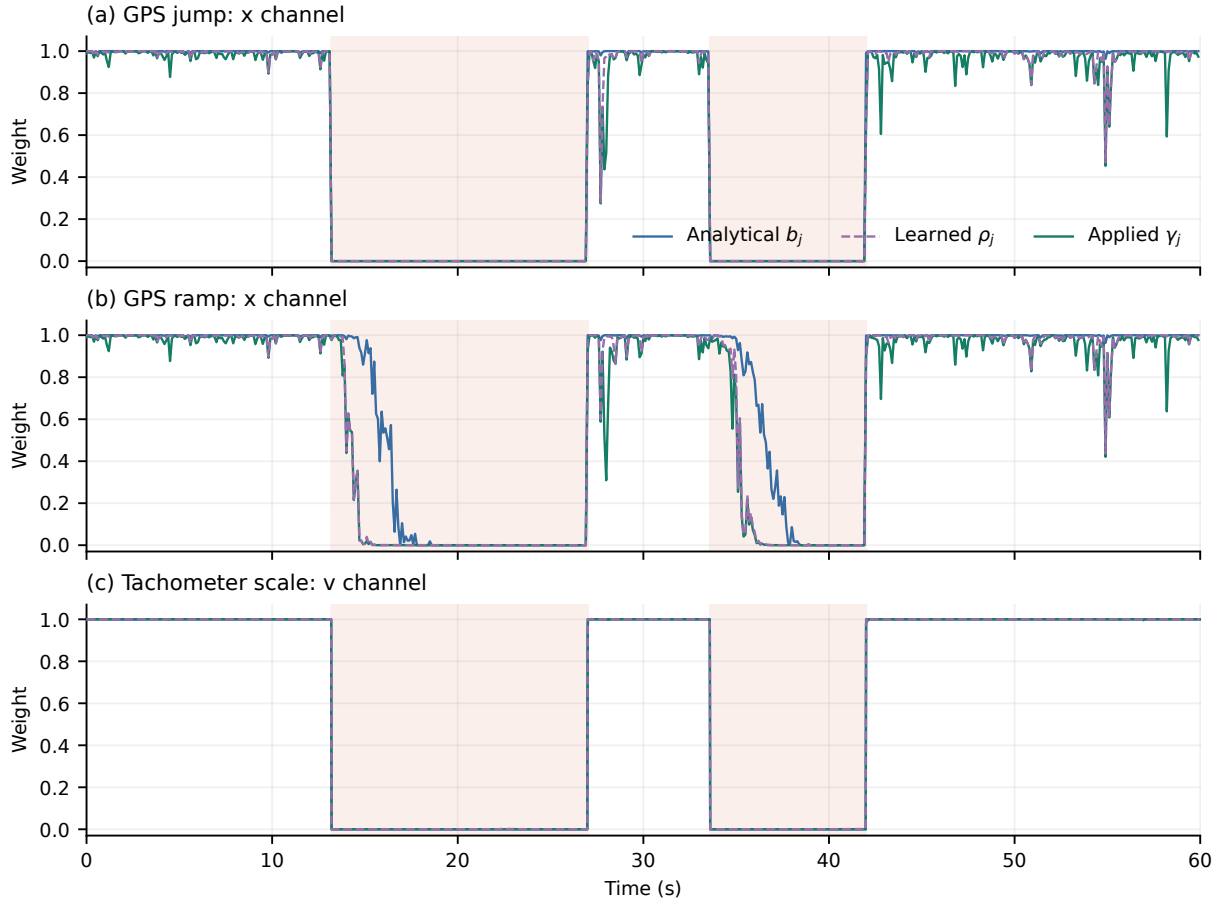


Figure 5: Trust decomposition for seed 101: GPS jump (x), GPS ramp (x), and tachometer scale corruption (v). Curves show analytical consistency b_j , learned reliability ρ_j and applied trust γ_j at available measurements. Position trust includes the minimum across both coordinates. Shading marks externally known injection intervals; it is not a filter input. The analytical factor already strongly rejects large jumps.

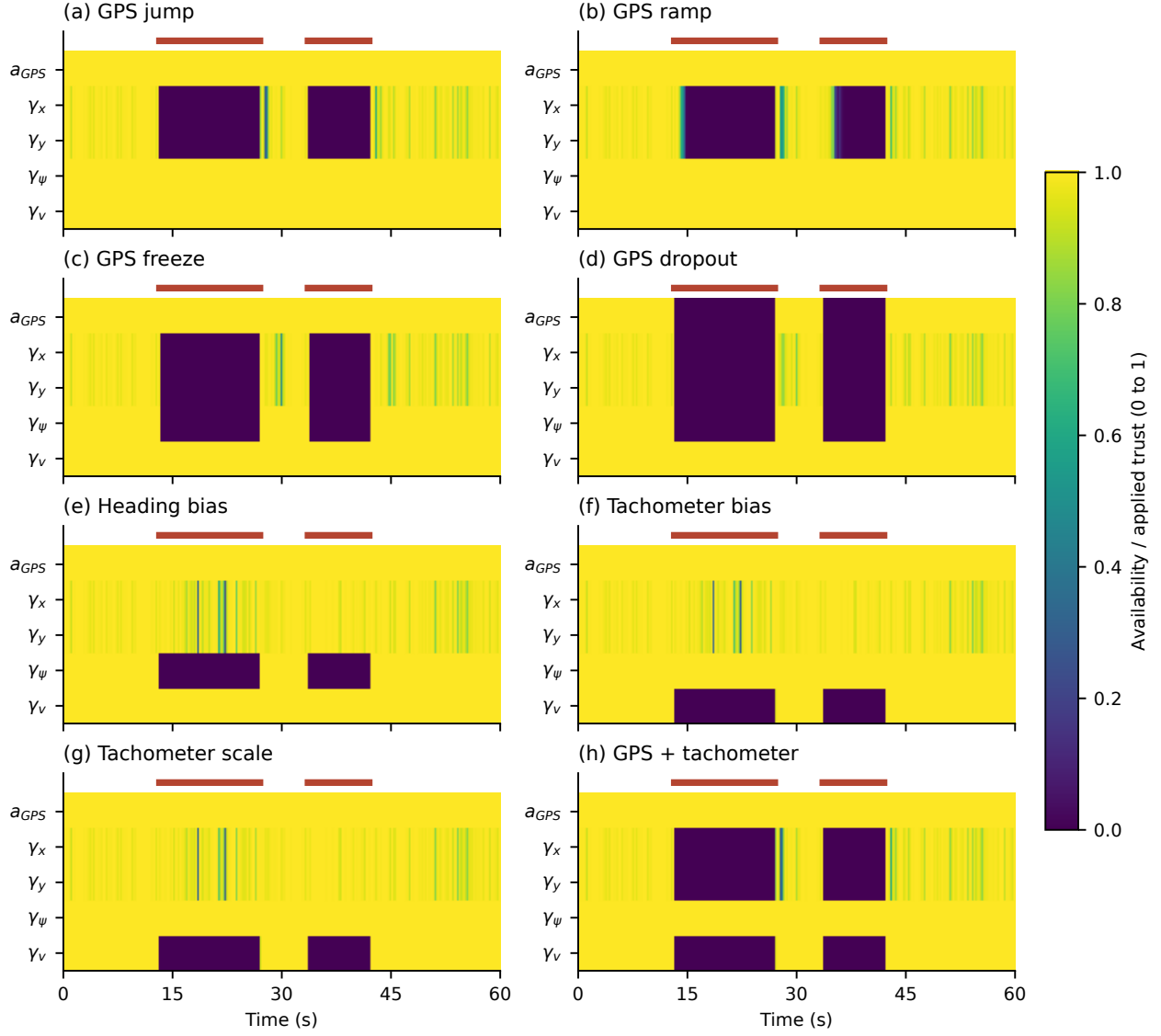


Figure 6: Applied mask outputs for eight fault conditions on seed 101. Each panel has one availability row and four trust rows, all on a common $[0, 1]$ scale. Dark colors mean unavailable or suppressed; bright colors mean available or highly trusted. Red bars above the panels identify injection intervals only for interpretation. GPS slots are shown at 10 Hz, including dropouts; velocity trust is subsampled for this display. The first row distinguishes missing GPS from valid but rejected GPS.

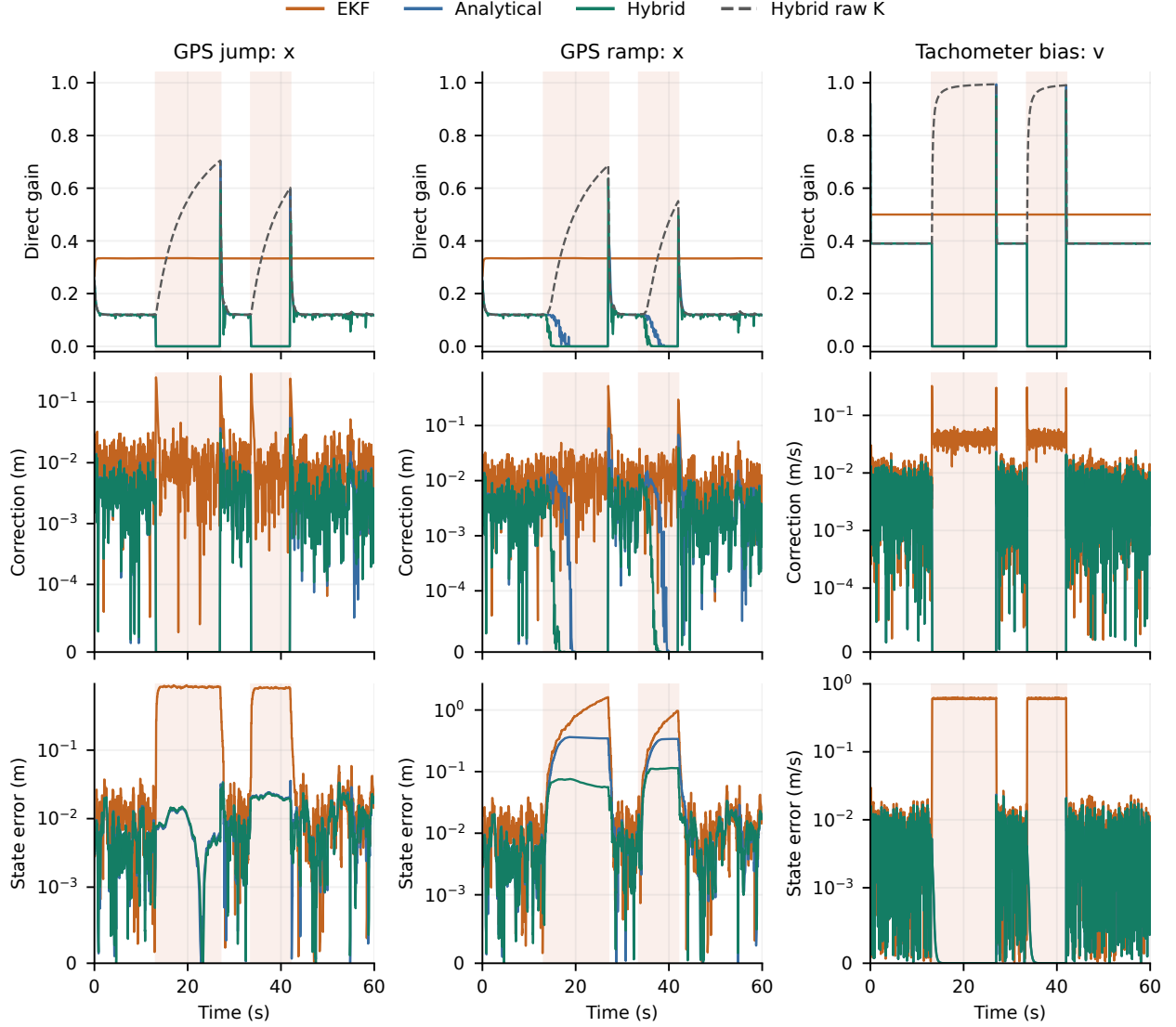


Figure 7: From gain to correction to error, seed 101. Columns show GPS jump (x), GPS ramp (x) and tachometer bias (v). Top: diagonal correction sensitivity for each method; the dashed curve is the hybrid's raw K_j , and its solid green curve is the applied $K_j\gamma_j$. Middle: magnitude of the complete measurement-induced state correction. Bottom: absolute state error against truth. GPS gain/correction curves use available GPS times; velocity curves use every step. Correction and error axes are symmetric-log scales, with linear regions below 10^{-4} and 10^{-3} respectively in the indicated units. Shading identifies injection intervals.

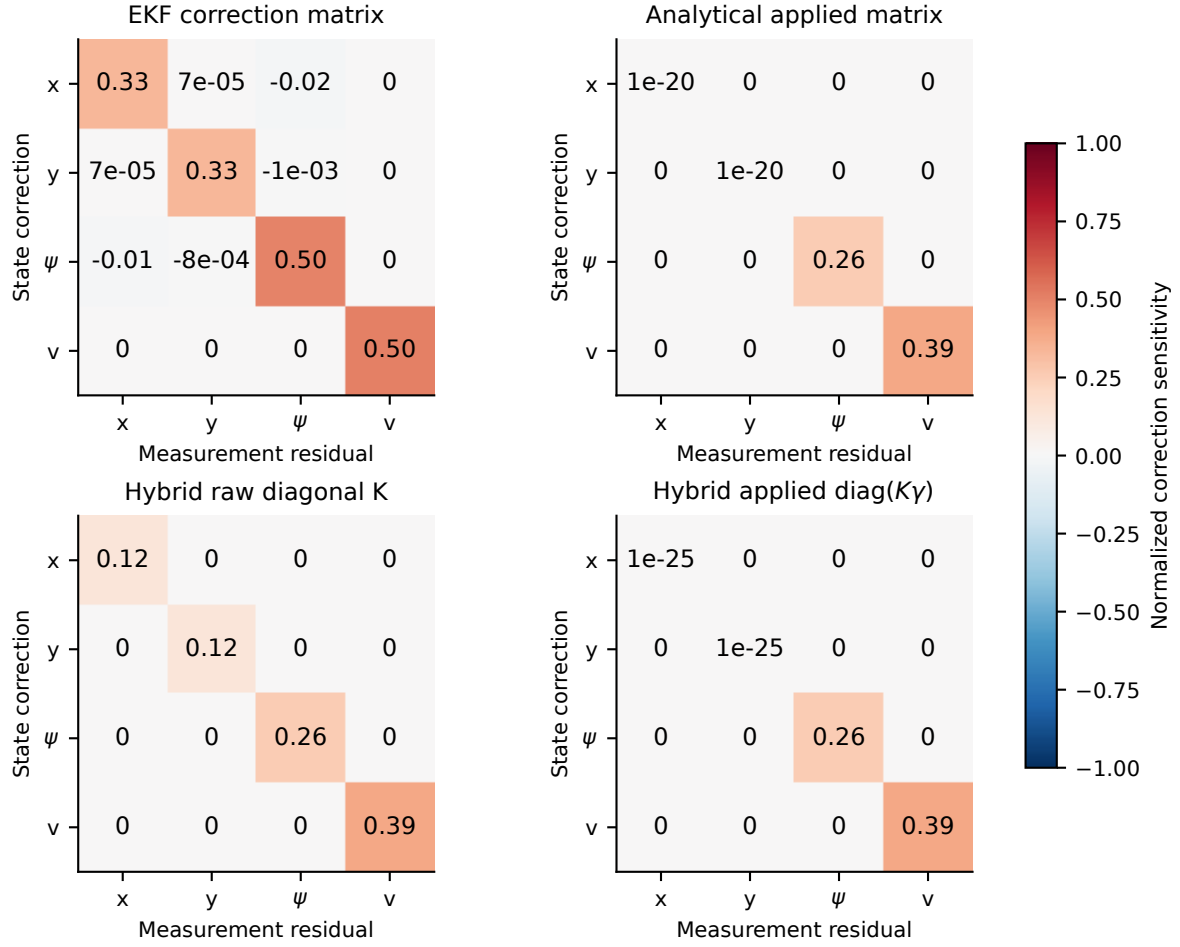


Figure 8: Correction matrices at the first attacked GPS-jump sample, $t = 13.2$ s, seed 101. Rows are state corrections and columns are measurement residuals, normalized by the stated fixed scales. Upper panels show the EKF and analytical ablation’s applied matrices. Lower panels separate the hybrid’s raw diagonal gain from its mask-modulated gain. Structural zero entries are labeled zero; tiny nonzero entries use scientific notation. This is one diagnostic instant, not an average or a matrix-norm performance metric.

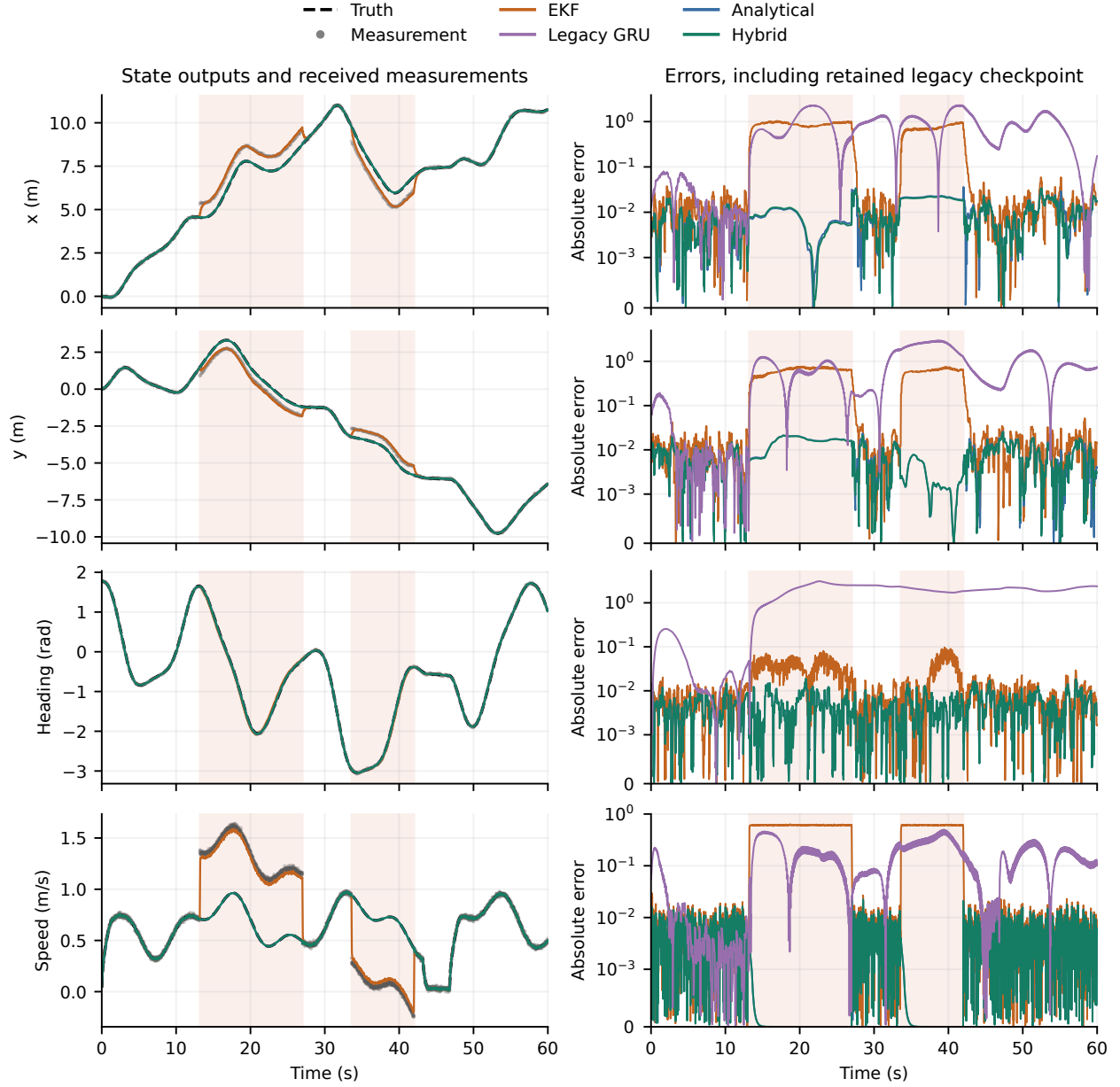


Figure 9: Observer outputs for combined GPS/tachometer corruption, seed 101. Left: x , y , heading and speed, with true state (black dashed), corrupted measurements (gray points) and three observer outputs. Right: absolute errors, including the legacy GRU. Heading curves are expressed on the truth's unwrapped branch for readability; heading errors are wrapped. Error axes use a symmetric-log scale with linear region below 10^{-3} in each row's units. Shading identifies the externally defined attacks.

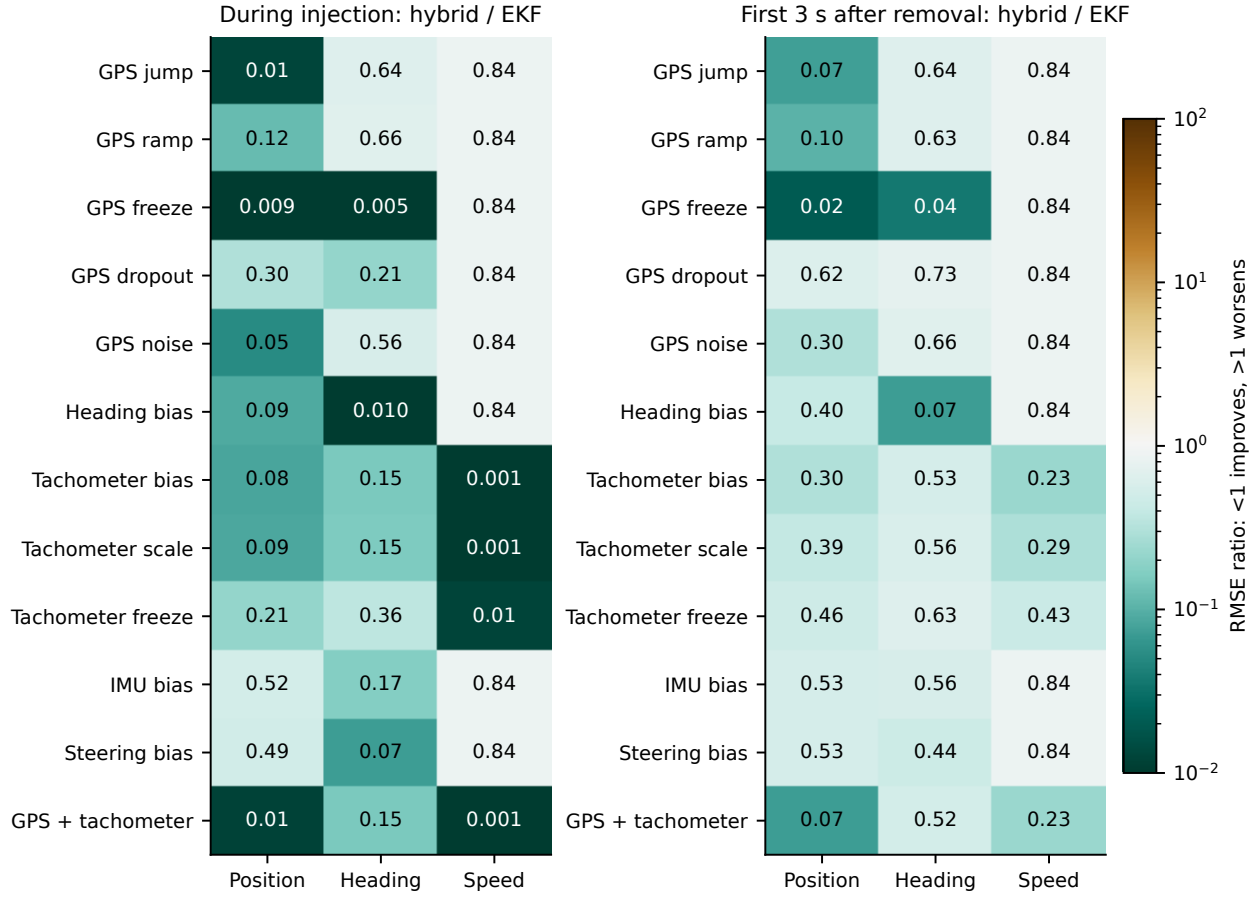


Figure 10: Hybrid-to-EKF pooled RMSE ratios during injection (left) and in fixed 3 s post-attack windows (right), for all 12 attacked nominal conditions and three seeds. The clean condition is excluded because it has no injection/removal events. Ratios below one favor the hybrid. Colors saturate outside $[0.01, 100]$; numerical labels give the values. A post-attack RMSE is not a settling-time estimate and can retain effects of earlier state contamination.

6.5 Closed-loop driving

A separate benchmark uses the repository Stanley lateral controller and PID speed controller. Each estimator drives its own MockQCar trajectory on a circle of radius 2 m with target speed 0.65 m/s. The vehicle starts at $(2, 0, \pi/2)$, at rest. Three seeds (401–403) run for 40 s at 50 Hz with 10 Hz noisy GPS. The nine conditions in Table 6 use an attack interval $[10, 24]$ s. Fault definitions match Table 3 except that the GPS jump is fixed to $(1, -0.7)$ m and applicable biases/ramps have positive sign. Controller settings are recorded in Appendix A.

Path error is the absolute radial deviation $|\sqrt{x^2 + y^2} - 2|$. Path and true-speed tracking RMSE exclude the first 3 s. This radial metric measures departure from the reference circle, not along-track timing or completion of an identical mission. Both path and speed must therefore be inspected.

Table 6: Closed-loop radial path RMSE (m) and true-speed tracking RMSE (m/s), pooled over three drives per condition after startup.

Condition	Path: EKF	Hybrid	Speed: EKF	Hybrid
Clean	0.0232	0.0227	0.0047	0.0057
GPS jump	0.4883	0.0249	0.0047	0.0057
GPS ramp	0.4388	0.0227	0.0047	0.0057
GPS freeze	1.7315	0.0231	0.0047	0.0057
GPS dropout	0.0257	0.0237	0.0047	0.0057
Heading bias	0.5005	0.0217	0.0047	0.0057
Tachometer bias	0.0194	0.0228	0.3651	0.0056
Tachometer freeze	0.0240	0.0228	0.0544	0.0057
GPS + tachometer	0.0465	0.0253	0.3651	0.0056

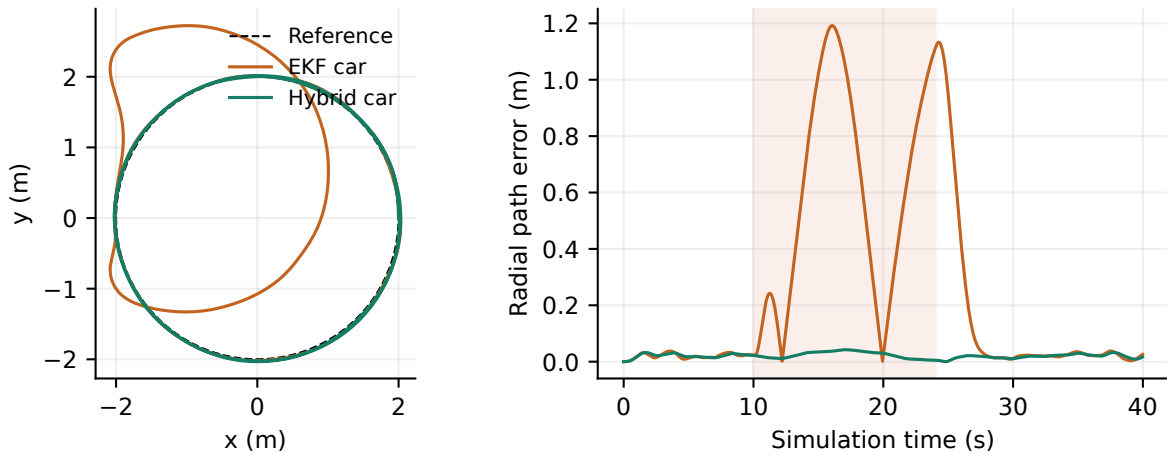


Figure 11: GPS-jump driving example, seed 401. Left: separate true plant trajectories under each estimator. Right: radial path error; shading marks the attack interval. These are controlled-plant outputs, not two estimates of one common trajectory.

With a GPS jump, path RMSE decreases from 0.4883 m for the EKF-controlled car to 0.0249 m for the hybrid-controlled car. Under heading bias it decreases from 0.5005 to 0.0217 m. Not every path metric improves: under tachometer bias the EKF has slightly lower radial error, 0.0194 versus 0.0228 m, but much worse speed tracking, 0.3651 versus 0.0056 m/s. That case illustrates why a single path metric can misrepresent driving performance. This test exercises the estimator/controller/plant loop headlessly; it does not validate the entire networked Ground Station execution stack.

6.6 Calibration mismatch and a high-rate check

A stress test uses seeds 201–202, 60 s drives, 5 Hz GPS, a plant time constant increased by 20%, and a steady-state speed curve increased by 10%. The estimator retains nominal calibration. Table 7 shows the harder combined-fault case explicitly.

Under simultaneous GPS and tachometer corruption, the hybrid gives 0.4638 m, below the EKF’s 0.6571 m but above the ablation’s 0.4019 m. Rejecting the absolute measurement leaves propagation dependent on

Table 7: Position RMSE (m) under plant-only calibration mismatch and 5 Hz GPS, pooled over two drives. The analytical ablation outperforms the hybrid in the combined attack.

Condition	EKF	Analytical	Hybrid
Clean	0.0265	0.0163	0.0158
GPS ramp	0.5919	0.1816	0.0709
GPS dropout	0.0659	0.0210	0.0201
GPS + tachometer	0.6571	0.4019	0.4638

an imperfect model; stronger rejection need not minimize total trajectory error. This limits claims of universal benefit from learning.

Figure 12 places the complete closed-loop comparison beside the calibration-mismatch experiment. It retains cases where the hybrid is worse on a particular metric. In particular, the tachometer-bias path result and combined-fault stress result should be considered alongside the gains in speed tracking and nominal localization, rather than omitted from a visual robustness argument.

A separate one-seed check (203) uses 200 Hz estimation and GPS, 15 s drives, no added sensor noise, and matched calibration. For GPS jump and dropout, hybrid position RMSE is approximately 8.54×10^{-5} and 1.37×10^{-4} m. These small values characterize a favorable near-matched simulation regime. They are not evidence of corresponding physical-car localization accuracy.

6.7 Computational cost

Runtime evaluates the exported MLP with NumPy on CPU. Table 8 summarizes saved nominal timing observations. For each estimator it reports the median of 39 per-drive medians and the median of 39 per-drive 95th percentiles; the latter is not a pooled p95 or a worst-case bound.

Table 8: Saved estimator-update timing in the nominal benchmark.

Method	Median trial median (ms)	Median trial p95 (ms)
EKF	0.405	0.670
Legacy GRU	6.792	8.471
Analytical	0.631	0.992
Hybrid	0.769	1.235

The hybrid is substantially cheaper than the retained GRU in this run, but slower than the EKF. Measurements exclude the GUI, network bridge and full control-stack scheduling. The original timing files do not contain a complete machine specification or load record, so these values are descriptive rather than portable latency guarantees. The high-rate check likewise does not establish a hard real-time deadline.

7 Discussion and limitations

The experiments support a practical division of responsibility. Calibrated propagation and explicit guards handle several large or readily recognizable faults; learned trust adds discrimination in cases such as gradual GPS displacement and excessive GPS noise. Attributing the full EKF-to-hybrid improvement to the neural network would contradict the ablation results.

Several limitations remain. First, the physical model and rejection scales are tuned to the simulated QCar operating range. Sustained wheel slip, unexpected actuator dynamics or noisy steering measurements can make correct observations look inconsistent. Second, the auxiliary odometry reference can drift and can be indirectly influenced by previous accepted corrections. It provides no independent absolute localization guarantee. Third, the covariance-like recursion is heuristic; uncertainty calibration and probabilistic coverage have not been measured.

The method suppresses unreliable information rather than reconstructing the attack signal or identifying an attacker. Coordinated corruption that remains consistent with the motion model may evade rejection. The available experiments use synthetic, predefined faults, modest numbers of drives and known initial poses. The fault set is shared between training and test, and the comparisons do not include a separately tuned

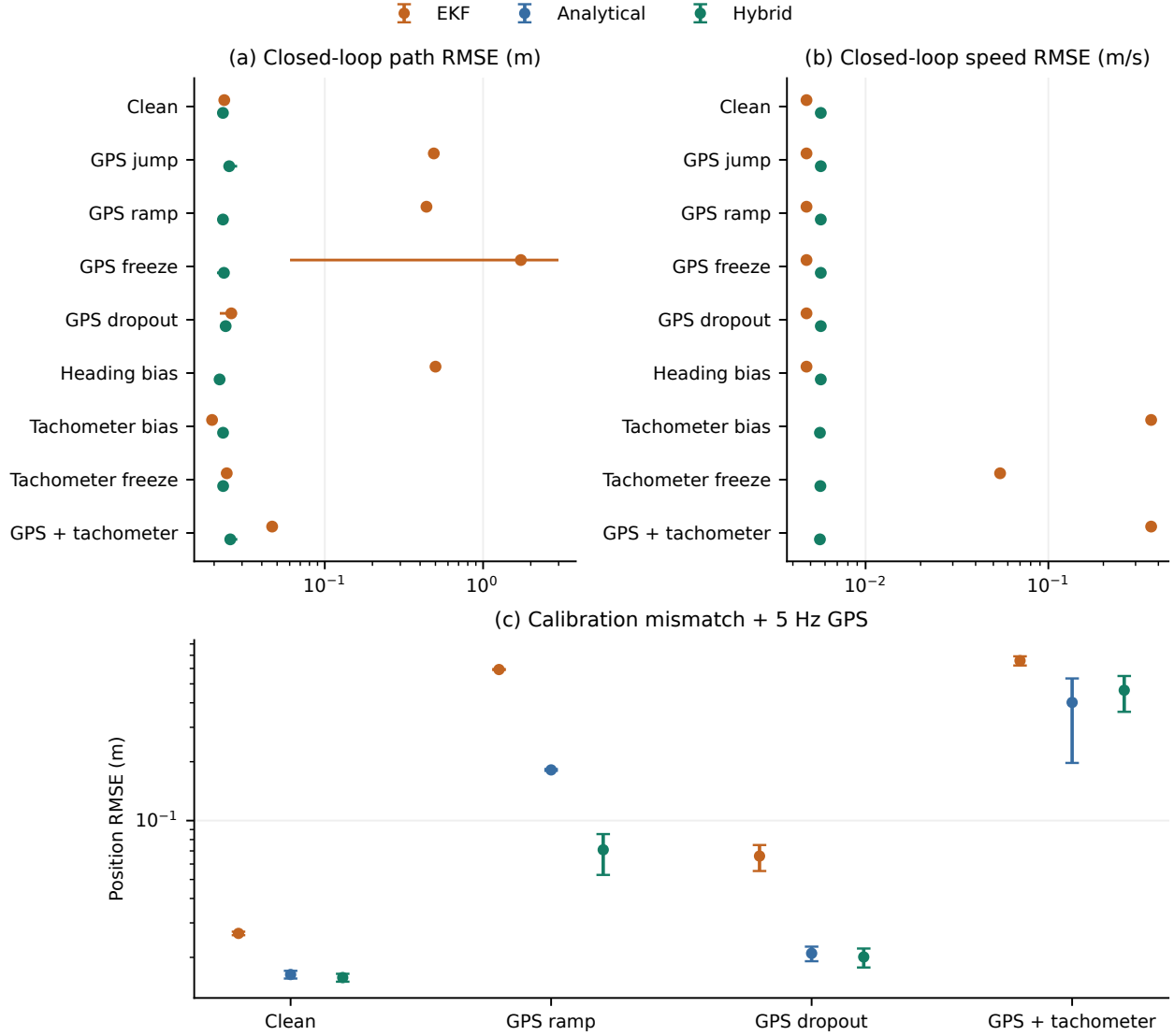


Figure 12: Driving performance and model-mismatch limits. (a,b) Closed-loop path and speed tracking RMSE for all nine tested conditions, pooling three separate plant trajectories per estimator and excluding startup. Dots show pooled RMSE and segments span per-drive values. (c) Position RMSE under calibration mismatch and 5 Hz GPS, with two seeds; error bars span those two values. All error axes are logarithmic. The analytical ablation has lower combined-attack stress RMSE than the hybrid. Ranges describe these observed drives and are not confidence intervals.

innovation-gated or robust-loss EKF. Such baselines are needed before claiming an advantage over robust filtering generally.

No physical-car trials are reported here. The legacy GRU comparison is a checkpoint comparison, not a controlled architecture comparison after equal retraining. The study also lacks separate ablations for the anchor, individual guards and loss terms. These limitations motivate tests with independently calibrated plants, unseen attack mechanisms, additional robust baselines and physical sensor logs with an independent position reference.

8 Conclusion

The revised RobustKLNet simulation backend combines physical prediction, analytical consistency gates and learned measurement trust with diagonal correction gains. It uses no online attack labels or clean sensor reference. Continuous simulated estimation and closed-loop driving improve over the tested repository EKF across many injected faults. The neural ablation shows where learning helps and where the explicit analytical design already accounts for most of the robustness. The observed failure under combined corruption and calibration mismatch remains part of the result. The present evidence supports this hybrid implementation for the specified simulation tests; broader robustness and transfer require further evaluation.

Reproducibility and evidence

All reported primary metrics come from saved benchmark JSON/NPZ pairs in `Robust/results: heldout, closed_loop, stress_mismatch`, and `native_rate`. The checkpoint is `models/innovation_trust_sim.npz`. Its SHA-256 fingerprint is

```
42fc0835b0f22c2f777ad7a9557ec2e0925
c85c5ddec1b7b8a289fb0d6851478
```

All four result files record that fingerprint. The asset builder pools metrics directly from the saved per-drive rows, checks checkpoint identity, and verifies 27 diagnostic rollouts (nine conditions, three observers, seed 101) against saved trajectories to absolute tolerance 10^{-7} . Instrumentation observes the EKF’s existing gain-reporting hook without replacing its update equations. For every diagnostic step, the recorded prediction plus complete gain-residual correction must reconstruct the returned state (including the runtime adapter’s float32 publication), and reconstructed availability/consistency/network factors must match the applied mask. The diagnostics and source fingerprints are retained in `paper_assets/observer_diagnostics.npz` and its matching manifest. Derived attack/post-attack metrics are saved in `phase_metrics.json`; full-drive comparison ratios are in `relative_metrics.json`, in the same asset directory. The expanded figure sets are exported as vector PDFs and reusable PNG images. Source hashes are recorded in `paper_assets/evidence_manifest.json`. This verifies artifact consistency, not historical provenance of every code revision. The old mixed GRU/revision report is retained separately in `archive/` and is not part of the current method description.

The dataset generator, training script, filter and benchmark scripts are included in the project workspace. No public repository URL or public dataset release is asserted. From the `Robust` directory, the relevant PowerShell commands are:

```
python train_robust_kalmanet.py --simulation-trust --epochs 60 --rounds 2
$ckpt = 'models/innovation_trust_sim.npz'
python validate_robust_kalmanet.py --simulation --duration 60 --legacy --checkpoint $ckpt
python closed_loop_benchmark.py
python -m unittest test_innovation_trust -v
```

The validation command writes to `results/simulation_metrics.json` by default; the reported original run was saved as `results/heldout.json`. Retraining and these output paths can replace associated local artifacts; a new checkpoint must be identified by its own fingerprint. The root-level `robust_workflow.cmd` creates separate run folders and supports headless evaluation, training and Ground Station web replay. Replay displays completed benchmark trajectories; it does not alter the metrics or stream the benchmark during computation. To rebuild this manuscript, run `build_paper_assets.py` from its folder and then run `pdflatex` twice on `robust_kalmanet_trust_report.tex`. Assets require the adjacent `../results` and `../models` directories.

A Numerical settings

Nominal throttle/speed interpolation uses

$$\begin{aligned}\mathcal{U} &= [0, .04, .06, .08, .10, .12, .14, .16], \\ \mathcal{V} &= [0, .21971, .35357, .49205, .62316, .75836, .89438, 1.02785] \text{ m/s}.\end{aligned}$$

Values are rounded here; the checkpoint and calibration YAML retain full precision. The gain recursion uses

$$\sqrt{\mathbf{P}_0} = [.02, .02, .02, .04], \quad \sqrt{\mathbf{Q}} = [.002, .002, .002, .006], \quad \sqrt{\mathbf{R}} = [.035, .035, .015, .012],$$

with state units in the order $[x, y, \psi, v]$ and $\mathbf{Q}$ referenced to a 0.02 s step. Valid estimator intervals satisfy $0 < \Delta t \leq 0.25$ s. For the controller test, Stanley uses $k_e = 2$, $k_{\text{soft}} = 0.2$, a 0.13 m position lookahead offset and a 0.5 rad steering limit. PID speed control uses $(k_p, k_i, k_d) = (0.12, 0.25, 0)$, feedforward gain 0.16 and maximum throttle 0.16. These are the evaluated settings, not optimized values for all controllers or vehicles.

B Deterministic guard details

The steering actuator state s_k^δ obeys

$$s_k^\delta = s_{k-1}^\delta + \Delta t_k \text{ clip}(10[\text{clip}(\delta_k, -.52, .52) - s_{k-1}^\delta], -3, 3), \quad (18)$$

and the bicycle rate is $\omega_k^{\text{bike}} = \bar{v}_k \tan[(s_{k-1}^\delta + s_k^\delta)/2]/L$. A non-finite steering sample retains the previous steering state.

A gyro fault latch is set when its successive finite values differ by more than 0.3 rad/s while successive steering signals differ by less than 0.05 rad. It is cleared when $|\omega_k^{\text{IMU}} - \omega_k^{\text{bike}}| < 0.12$ rad/s. Subject to a clear latch and a finite gyro, the gyro is accepted if either its discrepancy from the bicycle rate is below 0.3 rad/s or $|a_k^{\text{IMU}} - a_k^{\text{model}}| < 0.65$ m/s². Otherwise the bicycle rate is used. The acceleration consistency term uses the acceleration returned by propagating the command-driven auxiliary speed. These are implementation rules; an attacker is not assumed to announce a gyro fault.

The tachometer freeze timer accumulates Δt_k only while successive values are exactly equal. Repetition for more than 0.15 s is rejected when $\max(|v_k^{\text{tach}}|, |v_k^{\text{cmd}}|) > 0.05$ m/s. The feature-level repeated-value accumulator instead uses a difference threshold of 10^{-7} and increments by the current interval since the last available GPS update, lower-bounded by Δt_k , for each available channel. It is therefore not a uniformly sampled physical freeze duration for all channels. If both GPS position accumulators exceed 0.15 and $|v_k^{\text{use}}| > 0.08$ m/s, the analytical terms for GPS position and GPS heading are zeroed. This exact-freeze policy is suited to repeated packets in the tested simulator and requires reassessment for quantized or noisy real sensors.

Availability follows per-channel finite checks and supplied position/heading validity flags. It is not inferred from the simulator’s attack indicator. Between GPS packets, direct position and heading corrections are absent. The main state, auxiliary reference, previous observations, discrepancy statistics, timers and gyro latch are reset once at initialization of each drive.

References

- [1] G. Revach, N. Shlezinger, X. Ni, A. López Escoriza, R. J. G. van Sloun, and Y. C. Eldar, KalmanNet: Neural network aided Kalman filtering for partially known dynamics, *IEEE Transactions on Signal Processing*, vol. 70, pp. 1532–1547, 2022. doi:10.1109/TSP.2022.3158588.
- [2] P. Dahal, S. Mentasti, L. Paparusso, S. Arrigoni, and F. Braghin, RobustStateNet: Robust ego vehicle state estimation for Autonomous Driving, *Robotics and Autonomous Systems*, vol. 172, article 104585, 2024. doi:10.1016/j.robot.2023.104585.